

P1

End-to-End Big Data Pipeline

FOR AMAZON REVIEW SENTIMENT & RATING PREDICTION

KIRSTEN LIVINGSTON

DSC400-T301 BIG DATA, TECHNOLOGY, & ALGO (2263-1) WINTER 2026

PROFESSOR NASHEB ISMAILY

[{LINK TO PROJ. SITE}](#)

[{LINK TO PRESENTATION}](#)

[{LINK TO GITHUB}](#)

Abstract: This report presents a comprehensive analysis of Amazon Customer Reviews, designed to extract insights from [Amazon_dataset 2026](#) (~123 GB) created by [Sahitya Sahu on Kaggle](#). The dataset contains millions of customer reviews, ratings, product metadata, and timestamps, many of which often contain unstructured text, missing values, and class imbalances, which complicate predictive modeling. By leveraging NLTK for text preprocessing, feature engineering, and fine-tuning DistilBERT for sentiment classification, the pipeline achieved an 82.6% accuracy on review sentiment prediction. This project's primary goal is to demonstrate a scalable framework for converting raw review data into actionable insights.

Introduction	1
Business problem & Hypothesis	1
Methodology & Technical Analysis	1
Phase 1 Data Ingestion & Integrity Auditing	1
Phase 2 Statistical Analysis & Feature Engineering	1
Phase 3 Machine Learning & NLP Modeling	1
Results	1
Recommendations and Ethical Considerations	1
Conclusion.....	1
References.....	1
Appendix A Data Quality (DQI) Framework	1
Appendix B Pipeline Matrix & Workflow	1
Appendix C Visualizations & Outputs.....	1
Appendix D Research Questions	1

INTRODUCTION

Amazon is one of the world's largest online marketplaces; they constantly generate millions of customer reviews across a wide range of products. These reviews provide valuable insights about customer preferences and incoming feedback that Amazon can utilize to better customer satisfaction and business strategy. However, the volume and unstructured nature of review data can create challenges for extracting meaningful insights at such scale.

This project analyzes Amazon customer review data using statistical analysis, machine learning, and transformer-based natural language processing techniques to identify patterns in customer sentiment and predict product ratings.

BUSINESS PROBLEM & HYPOTHESIS

Because Amazon relies heavily on customer feedback to better inform them about positive and negative business actions, it is vital to transform raw review data into actionable decisions. Although without automated analysis, valuable signals related within the data may be overlooked due to the sheer volume and complexity of customer reviews. By deploying an end-to-end data analysis pipeline, this project identifies and model patterns across structures features and unstructured textual review data.

Systematic analysis of Amazon customer reviews will:

- Reveal consistent sentiment patterns aligned with star ratings
- Predict product rating categories using structured and unstructured features
- Identify actionable insights to improve product performance

METHODOLOGY & TECHNICAL ANALYSIS

The data used for this project was sourced from Kaggle by Sahitya Sahu Amazon Large Dataset 2026 (~123GB), that is publicly accessible. Due to computational constraints, modeling was performed on a clean and feature-engineered subset designed to mimic production-scale workflows while maintaining reproducibility. The technical implementation followed a structured, phase-based analytical workflow to ensure reproducibility and validation across each stage of development.

PHASE 1 | DATA INGESTION & INTEGRITY AUDITING

The first phase transformed the large-scale Amazon review dataset into a structured analytical asset optimized for modeling. Through sampling manageable chunks of the full Amazon review dataset (~123GB raw), the master dataset began with a manageable size of data (129,794 reviews, 13 columns).

→ Distributed Data Acquisition

- Due to the dataset's scale, ingestion was performed in controlled chunks using Pandas. This mitigated memory overflow risks while keeping schema consistency across the data.

→ Structural Standardization

- The pipeline conducted datatype normalization, missing value remediation, duplicate removal, and schema validation checks. This process created a clean analytical subset of 129,794 records.

→ Feature Construction

- To enrich predictive power, these new features were created: review length (character count), sentiment polarity score (NLTK), reviewer activity metrics, product aggregated statistics, and temporal indicators (year, month, and day of the week)

→ Integrity Audit | Leakage Detection

- During this early inspection, reviewer and product-level average ratings appeared to be highly predictive. However, correlation testing revealed that these features encoded direct information about the target variable. This was due to data leakage, meaning the model was memorizing averages rather than learning patterns.
- With early integrity auditing, it was possible to prevent artificial performance inflation (~99% accuracy).

PHASE 2 | STATISTICAL ANALYSIS & FEATURE ENGINEERING

Following ingestion, the dataset underwent structured exploratory analysis to establish a baseline understanding of behavior and relationships within the dataset (McKinney, 2022).

→ Class Distribution Analysis

- The rating distribution revealed severe imbalance (5-star reviews ~66%) with low and medium ratings unrepresented.

→ Sentiment Correlation Testing

- Found a strong positive correlation was identified between sentiment polarity and star rating. Text sentiment is the dominant predictive signal in consumer review data.

→ Verified Purchase Hypothesis Testing

- T-tests and Chi-square tests revealed statistically significant differences between verified and non-verified reviews.

→ Distributional Observations

- Observed that most products receive few reviews; a small subset dominates volume, and most reviewers contribute minimally.

PHASE 3 | MACHINE LEARNING & NLP MODELING

After statistical validation, the project transitioned to predictive modeling. Multiple architectures were tested to diagnose performance constraints.

→ Model 1 — Baseline Logistic Regression

- Features: Basic numeric + Sentiment polarity
- Accuracy: 61%, Macro F1: 0.45
- The model heavily favored the dominate (high) class; medium ratings exhibited substantial overlap with both low and high, and minority recall was poor.

→ Model 2 — Feature-Enriched Model (Leakage Identified)

- Accuracy: ~99%, Macro F1: 0.98
- The model memorized reviewer and product average rating, while the performance looked amazing; it failed due to leakage contamination.

→ Model 3 — Clean Random Forest

- Accuracy: 77%, Macro F1: 0.58
- After removing leakage features, the model demonstrated stable generalization but highlighted that mid-range ratings are naturally harder to distinguish because they share many characters with both the low and high ratings.

→ Model 4 — Hierarchical Two-Stage Model

- Accuracy: 87%, Macro F1: 0.70
- This hierarchical architecture improved minority class performance by decomposing a complex multi-level class problem into sequential binary tasks. Where Macro F1 score increased from 0.58 to 0.70, demonstrating fairness across classes.

→ Transformer-Based NLP Validation (DistilBERT)

- Accuracy: 82.6%, Macro F1: 0.50
- Insights from this model show that context-aware embedding outperforms simple polarity scoring and demonstrates high classification accuracy even on reduced training subsets.

RESULTS

The analysis of the Amazon review dataset revealed many key patterns and model outcomes. Exploratory Data Analysis confirmed a significant class imbalance, with 5-star reviews making up ~66% of the dataset, while the low and medium ratings were underrepresented. Through sentiment analysis, a clear correlation between review text polarity and numerical ratings was identified, with more positive language associated with higher star ratings.

Statistical tests highlighted meaningful relationships between features like verified purchase status, having a strong association with high ratings, and sentiment polarity having the strongest predictor in regression models. These results demonstrate that the pipeline can extract meaningful patterns for review data while highlighting common challenges like class balance and mid-range rating ambiguity (Maas et al., 2011). Visualizations supporting these conclusions are found in Appendix C.

RECOMMENDATIONS AND ETHICAL CONSIDERATIONS

To ensure long-term reliability, the deployed model should be continuously monitored for performance drift and detection bias. Macro F1, class-level recall, and false positive/false negative rates should be tracked over time to identify disproportionate misclassification of minority rating classes. Also, periodic retraining using recent data and time-based validation splits should be implemented to maintain model robustness as customer behavior and product patterns evolve with time.

This project's dataset was sourced from a publicly available Kaggle repository (Sahitya Sahu, 2026). Ethical responsibility extends beyond initial model development to ongoing output auditing. Continuous evaluation is necessary to ensure that automated classifications do not overlook negative feedback or unfairly impact certain sellers. Manual oversight should remain in downstream decision-making processes, and regular fairness audits, due to data imbalance or shifts in marketplace dynamics.

CONCLUSION

This project highlights the value of structured analysis on large-scale Amazon review data to extract actionable business insights. The final model successfully identified sentiment patterns, while exposed to challenges like class imbalance and mis-range rating ambiguity, illustrating the complexity of real-world review data. Insights from this analysis can inform product strategy, marketing decisions, and future enhancements to the automated review analysis pipeline, including real-time monitoring and scalable deployment.

Moving forward, rather than working on a 130k subset, implementing a fully distributed pipeline across the entire original ~123GB dataset would increase big data scalability. To take it a step further, in future phases, model deployment would turn this project into a scalable, production-ready business intelligence system.

REFERENCES

- Amazon_dataset 2026: 123 GB Amazon Dataset for Real-World Machine Learning, NLP, Data Engineer.* (n.d.). Kaggle. <https://www.kaggle.com/datasets/sahityasahu/amazon-review-dataset>
- Loper, Klein, Bird, E., Ewan, Steven. (2009, June 12). *Natural Language Processing with Python*. O'Reilly Online Learning. <https://www.oreilly.com/library/view/natural-language-processing/9780596803346/>
- Jurfsky, Martin, D., James. (2026, January 6). *Speech and language processing*. <https://web.stanford.edu/~jurafsky/slp3/>
- Maas, A., Daly, R. E., Pham, P. T., Huang, D., Ng, A. Y., & Potts, C. (2011, June 1). *Learning word vectors for sentiment analysis*. ACL Anthology. <https://aclanthology.org/P11-1015/>
- Kowsari, K., Jafari Meimandi, K., Heidarysafa, M., Mendu, S., Barnes, L., & Brown, D. (2019). Text Classification Algorithms: A Survey. *Information*, 10(4), 150. <https://doi.org/10.3390/info10040150>
- McKinney, W. (2022, August 15). *Python for Data Analysis, 3rd Edition*. O'Reilly Online Learning. <https://www.oreilly.com/library/view/python-for-data/9781098104023/>

APPENDIX A | DATA QUALITY (DQI) FRAMEWORK

Completeness:

All major aspects of the project were addressed; this includes data ingestion, cleaning, exploratory data analysis, feature engineering, sentiment and rating modeling, and visualization. Each step is documented in separate Jupiter Notebooks, and the final dashboards (both interactive and static in case of interactive link failure). The dataset used for this project was sourced from a publicly available Amazon Reviews dataset (~123GB) and processed using chunk-based ingestion to create a manageable master dataset.

Code Quality:

The notebook's code was structured using modular functions. Data cleaning steps were transparent, including handling missing values, converting data types, removing duplicates, and engineering key features. Advanced modeling workflows, including NLP preprocessing, were clearly separated into stages within the notebook. All notebook phases and code are highlighted in the GitHub Repository as well as the link to the original data source. This organization ensures that future users can replicate or potentially extend the project.

Analysis Quality

This analysis combines exploratory data analysis, statistical modeling, and machine learning to uncover valuable and actionable insights. Statistical analysis observed correlations and evaluated rating predictors. Visualization quality was ensured throughout both the static and interactive

dashboards. After final refinements, the final dashboards display clear, non-technical friendly visuals, KPIs, and explanations about the data.

Innovation and Creativity:

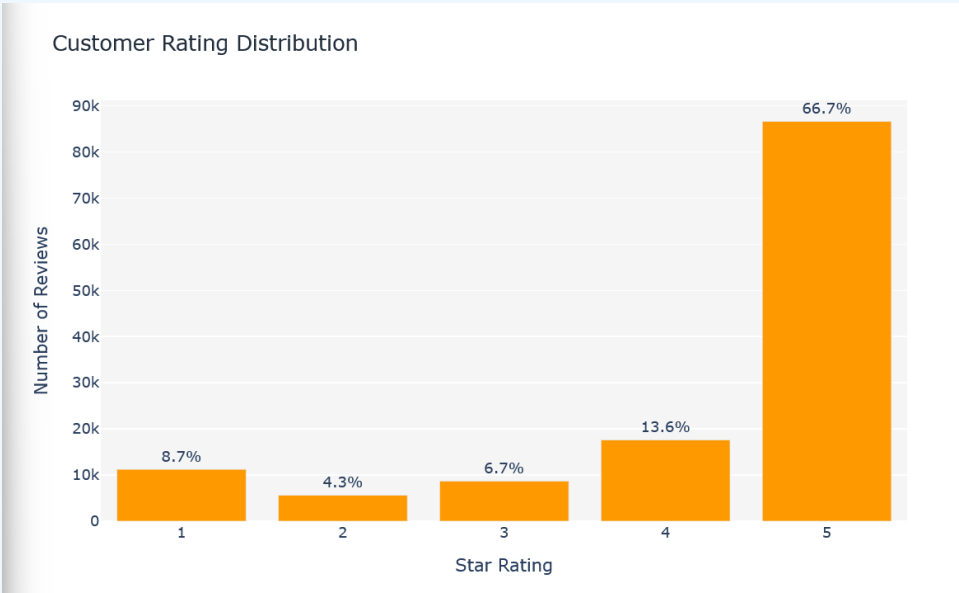
Innovation is demonstrated through the scalable handling of a large, real-world dataset, by using chunk-based ingestion to effectively process and sample a ~123GB dataset without overwhelming local memory resources. In addition, the development of both interactive and static dashboards enhances the insights and findings for this project. Although PySpark was not implemented due to system limitations, this project aims to adapt and fulfill an end-to-end analysis through optimized local processing strategies.

APPENDIX B | PIPELINE MATRIX & WORKFLOW

Phase	Step	Stack	Input	Output/Deliverable
Data Ingestion & Cleaning	Collect & Clean Amazon Reviews	Pandas, NumPy	Raw Dataset (123 GB)	Master Cleaned Subset (~130k rows)
EDA & Feature Engineering	Summary Stats, Distributions, Correlations, Create New Features	Pandas, Matplotlib, Seaborn	Cleaned Dataset	Engineered features, class imbalance identification
Visualizations	Exploratory Plots	Matplotlib, Seaborn	Cleaned & Engineered Dataset	Initial Plots and insights from data
Dashboards	Static & Interactive Dashboards	Plotly, HTML (GitHub Project Dashboard)	Cleaned Dataset & Key Findings	Static & Interactive Dashboards
Statistical Analysis	Modeling & Testing	Statsmodels	Feature-engineered dataset	Statistical validation of relationships, summaries
Modeling, Deep Learning, & NLP	Imbalance handling, NLP preprocessing	NLTK, Hugging Face Transformers, DistilBERT	Engineered features & tokenized review text	Trained ML models, sentiment classifier (` 82.6% accuracy)

APPENDIX C | VISUALIZATIONS & OUTPUTS

Figure 1 | Customer Rating Distribution



Breakdown: Illustrates class imbalance within the dataset, highlighting the sheer dominance of 5-star reviews following 4-star reviews (13.6%). 2-stars have the lowest count of reviews at 4.3%.

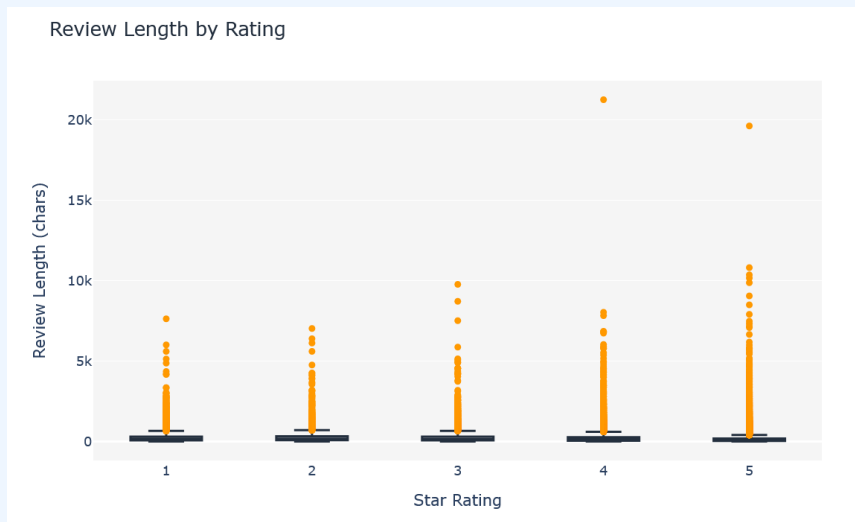
Figure 2 | Statsmodels Summary Table

OLS Regression Results						
=====						
Dep. Variable:	overall	R-squared:	0.202			
Model:	OLS	Adj. R-squared:	0.202			
Method:	Least Squares	F-statistic:	1.643e+04			
Date:	Wed, 18 Feb 2026	Prob (F-statistic):	0.00			
Time:	18:39:56	Log-Likelihood:	-2.0096e+05			
No. Observations:	129794	AIC:	4.019e+05			
Df Residuals:	129791	BIC:	4.020e+05			
Df Model:	2					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	3.7103	0.005	734.414	0.000	3.700	3.720
review_length	-7.32e-05	9.35e-06	-7.832	0.000	-9.15e-05	-5.49e-05
sentiment_polarity	1.7291	0.010	176.283	0.000	1.710	1.748
=====						
Omnibus:	20230.414	Durbin-Watson:	1.986			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	31079.278			
Skew:	-1.140	Prob(JB):	0.00			
Kurtosis:	3.740	Cond. No.	1.32e+03			
=====						

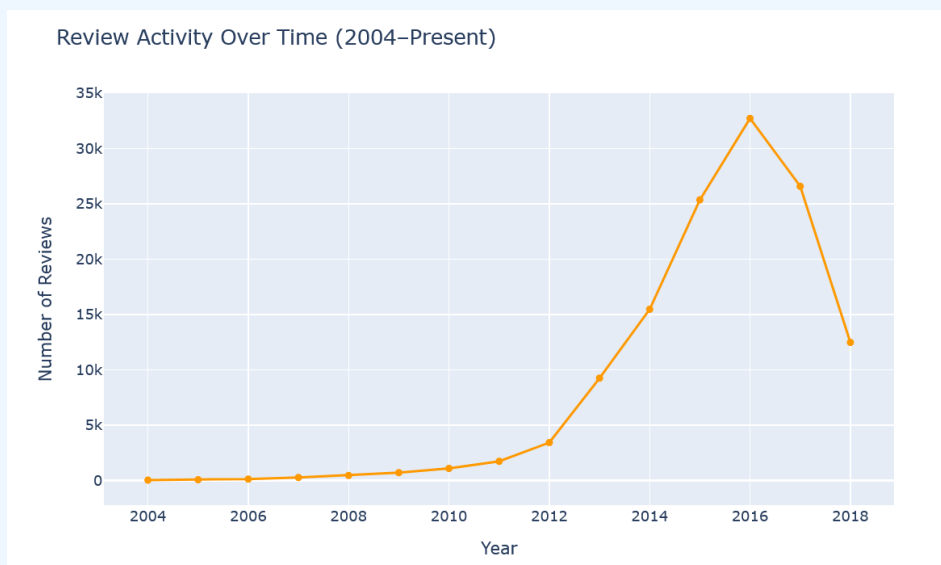
Breakdown: Contains statistical evidence that sentiment polarity significantly predicts star ratings, signifying feature engineering, and modeling decisions.

Figure 3 | Review Length by Rating



Breakdown: Visualizes how review length varies by rating where: 5-star reviews show a prominent long-review outlier, 4-star reviews have a smaller extreme, and 3-star reviews display many small-length outliers.

Figure 4 | Review Activity Over Time



Breakdown: Shows the volume of reviews submitted across years and months, detailing the seasonal trends and periods of high customer engagement. The highest peak reaches between 30k to 35k reviews, with a steady rise starting from 2004 to the drop in 2016.

APPENDIX D | RESEARCH QUESTIONS

Statistical Q & A

1. Do more positive reviews actually correspond to higher ratings?

Yes. Reviews that use more positive language usually have higher star ratings. This shows that the words people use in their reviews generally match the number of stars they give.

2. Do low-star reviews tend to be longer complaints?

Generally, yes. Reviews with fewer stars often include more detailed explanations or complaints, making them longer than higher-rated reviews.

3. Are some products consistently rated higher than others?

Not exclusively, the top products all tend to be positive reviews, there is not strong enough evidence that suggest some products are rated significantly higher than others. Differences in rating apply more about the individual and sentiment than the product itself.

4. Do the most reviewed products receive significantly different average ratings?

No. Products with many reviews usually maintain similar positive ratings to other popular products. This implies that the number of reviews does not strongly affect the overall rating.

5. Do verified purchase reviews differ in rating compared to non-verified reviews?

Yes. Verified purchases tend to have slightly different rating patterns compared to non-verified ones. This ensures that we are focusing on authentic experiences.

Modeling Q&A

6. Can review sentiment and metadata predict customer ratings at scale?

Yes. Reviews with more positive language and certain reviewers or purchase patterns tend to have higher ratings. This allows us to make accurate predictions of how a review will be rated.

7. Does a deep learning model outperform traditional ML by capturing richer text patterns?

Yes. Deep learning models, like fine-tuned DistilBERT, better understand the meaning in review text and subtle sentiment cues, which helped improve prediction accuracy compared to simpler models conducted in the modeling phase.

8. What features contribute most strongly to rating prediction?

The strongest predictors are sentiment subjectivity, sentence count, and average word length.

Positive language, verified purchases, and consistent reviewer behavior all tend to correspond to higher ratings.